CCCA Cell-State Analysis — Caron et al. 2020, Hematologic

Acute Lymphoblastic Leukemia · 10x · 25,784 malignant lymphoblasts (8 samples, 3 genetic subtypes)

GO / NO-GO GATES

Gate 1 — mixing & continuity

GO

Harmony (θ=10) fuses the three genetically distinct subtypes: same-sample neighbour fraction 1.00→0.26, iLISI 1.0→4.2/8 (sample) and 2.5/3 (subtype), one connected kNN component at n_pcs=20/30/40. Judged on the regenerated embedding across a θ-sweep — B-ALL (ETV6-RUNX1, HHD) and T-ALL intermix on one continuous manifold; the only within-manifold structure is a cycling↔quiescent axis (both poles carry all 3 subtypes), not a lineage split.

Gate 2 — module geography

GO (non-branching)

Signatures collapse onto a single (Gast/pSom)→Crest axis: smoothed pSom↔Crest r=−0.74, Gast↔pSom r=−0.46 (organized cross-repression), Gast↔Crest r=−0.20. Gast and pSom sit at OPPOSITE poles — no co-expression hub, no two competing arms. Crest is a terminal gradient (joint pSom&Crest enrichment 0.28 < 1). Along pseudotime Gast falls (ρ=−0.73), Crest rises (decile pt 0.68). Principal curve fold-free (2 tips / 0 forks, self-approach 2.97). Crest does NOT enrich a Gast arm → no flag.

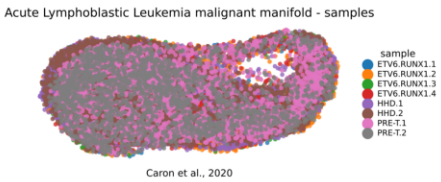
Gate 3 — hub cross-repression

NA

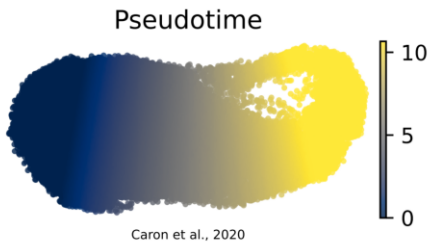
Non-branching topology — no bifurcation and no Gast/pSom hub, so the branching-only cross-repression test does not apply. Correlation & modules notebooks are correctly not produced.

STEP FIGURES

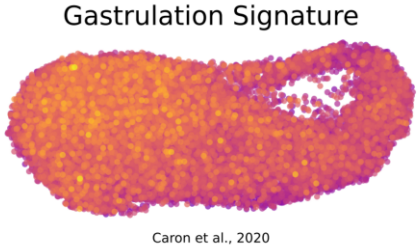
01 · Samples (integration)



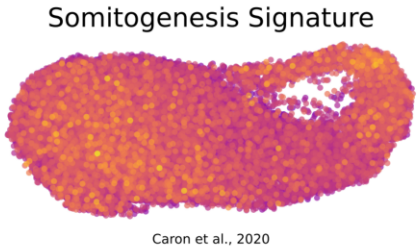
02 · Pseudotime (curve)



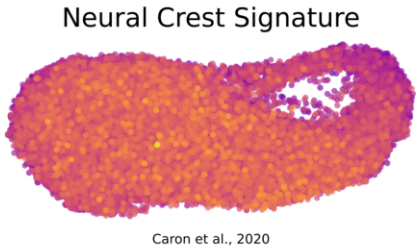
03 · Gastrulation



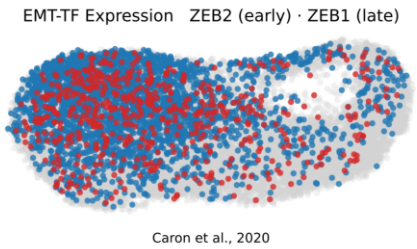
03 · Somitogenesis



03 · Neural Crest



04 · EMT-TFs (ZEB2→ZEB1)



Dataset: Pediatric ALL (Caron 2020), 10x — 25,784 malignant lymphoblasts over 8 samples spanning B-ALL (ETV6-RUNX1), B-ALL (HHD) and T-ALL.

Verdict: Pattern reproduces as the NON-BRANCHING case — one gastrulation/somitogenesis→neural-crest developmental axis with organized Gast↔pSom cross-repression; no hub, no bifurcation. G1 GO / G2 GO / G3 NA.

PARAMETER CHOICES & RATIONALE

Min-cell threshold	≥100 floor — counts fall smoothly (max ratio 1.52, top/second 1.16 → no dominant cliff); all 8 samples kept
Integration	Harmony on sample, θ=10 (strong; mixing plateaus θ=10–15, θ=20 regresses); n_pcs=30, n_neighbors=30, spread=1.3
Islands removed	4 cells / 3 detached specks (UMAP radius graph, largest component)
Geometry	Non-branching → scFates principal curve (ElPiGraph), Nodes=20, λ=0.3, μ=100; 2 tips / 0 forks, fold-free
Root node	Curve tip 0 — Gast/pSom-high, Crest-low (developmental origin; Gast→Crest along pseudotime)
Hub identity / r	None — non-branching, no Gast/pSom convergence hub (Gate 3 NA)
Module means (raw)	Gast −0.017 · pSom +0.000 · Crest −0.010 → Linear
EMT-TFs shown	ZEB2 (root, 40.6% expr) → ZEB1 (terminus, 8.9%) — only detectably-expressed distinct factors; SNAI2/TWIST1/FOXC2 <0.1%, gated out